GUJARAT TECHNOLOGICAL UNIVERSITY

Shree Swami Atmanand Saraswati Institute of Technology

A

Project Report on

(Energy Conservation)

Under subject of

Social Internship (BE05000011)

B.E 3rd year, semester – V

COMPUTER ENGINEERING

Under the Guidance of

(Prof. Pritesh Bhimani)

Submitted by group:

Sr. No.	Name	Enrollment No.
1	Yashvi Sureshbhai Vekariya	240760107203
2	Rensi Rajeshbhai Dungrani	240760107164
3	Itisha Maheshbhai Zalavadiya	240760107212
4	Vidhi Shaileshbhai Narola	240760107204
5	Twinkal Dilipbhai Variya	240760107197
6	Krisha Dipakbhai Vastarpara	240760107198
7	Rajvi Alpeshbhai Rupavatiya	240760107167

Academic year: 2026-2027

SHREE SWAMI ATMANAND SARASWAT IINSTITUTE OF TECHNOLOGY

Opp. Kapodra police station, varachha

road, Surat – 395006



CERTIFICATE

This is to certify that the Project entitled "Smart Community Energy Portal" has been prepared by Yashvi Sureshbhai Vekariya (240760107203), Rensi Rajeshbhai Dungrani (240760107164), Itisha Maheshbhai Zalavadiya (240760107212), Vidhi Shaileshbhai Narola (240760107204), Twinkal Dilipbhai Variya (240760107197), Krisha Dipakbhai Vastarpara (240760107198), and Rajvi Alpeshbhai Rupavatiya (240760107167) in the subject of Societal Internship (BE05000011) under my guidance in fulfillment of the degree of Bachelor of Engineering in Computer Engineering Department (5th Semester) of Gujarat Technological University, Ahmedabad during the academic year 2026-2027.

Date:

Place: SSASIT, Surat

Project Guide
Prof. Pritesh Bhimani

Head of Department
Prof. Chirag R. Patel

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We express our sincere gratitude to Gujarat Technological University (GTU) for providing us with the opportunity to participate in the Summer Internship Programme 2026. This internship has helped us enhance our technical knowledge, teamwork, communication, and problem-solving skills through practical learning.

We are thankful to the Principal, Head of the Computer Engineering Department, Internship Coordinator, and our Faculty Guide Prof. Pritesh Bhimani of Shree Swami Atmanand Saraswati Institute of Technology (SSASIT), Surat, for their valuable guidance, encouragement, and continuous support throughout the internship.

We also extend our gratitude to all the community members who participated in our energy conservation survey. Their responses and suggestions played an important role in the successful completion of this project.

Finally, we thank all our team members for their dedication, cooperation, and sincere efforts in completing the Smart Community Energy Portal project successfully within the given internship duration.

Abstract

The Smart Community Energy Portal was developed as a part of the Gujarat Technological University (GTU) Summer Internship Programme 2026 by a team of seven students from the Computer Engineering Department of SSASIT, Surat. The background of this project lies in the rising demand for electricity and the consequent need to promote responsible energy usage at the household and community level.

The objective of the project was to spread awareness about energy conservation by developing an informative website and conducting a community survey. The methodology adopted involved literature review, project planning, survey design using Google Forms, data collection from 100 respondents, data analysis using charts and graphs, and development of the website using HTML, CSS, and JavaScript.

The key findings revealed that 84% of respondents agree that energy conservation helps reduce environmental pollution, 71% regularly use natural light, 64% use LED bulbs, and 56% have observed a rise in electricity bills. Excessive use of air conditioners and fans (48%) was identified as the leading cause of household energy wastage.

Based on these findings, practical solutions and recommendations were proposed covering cooling energy management, behavioural change, complete LED transition, and expanded awareness programmes. The project successfully achieved its objective of promoting energy conservation awareness while demonstrating how Computer Engineering students can contribute to solving community-based challenges using digital solutions.

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Chapter 1: Introduction

1.1 Background

Energy is one of the most important resources used in daily life. It is required in homes, industries, schools, hospitals, transportation, and many other sectors. With the increasing use of electrical appliances and modern technology, the demand for electricity is also increasing every year. Therefore, using energy wisely has become very important.

Energy conservation means using electrical energy efficiently and avoiding unnecessary wastage. Simple habits like switching off lights and fans when not in use, using LED bulbs, and choosing energy-efficient appliances can help save electricity. These small efforts not only reduce electricity bills but also help protect natural resources and reduce environmental pollution.

Many people know that saving energy is important, but they are not always aware of the simple methods that can be followed in daily life. Creating awareness through campaigns, educational programs, and digital platforms encourages people to adopt energy-saving habits. Collecting data through surveys also helps in understanding how people use electricity and what challenges they face in conserving energy.

1.2 Objectives

Primary Objectives:

- To develop a user-friendly web portal on energy conservation.
- To create awareness about efficient use of electrical energy.
- To conduct a community survey using Google Forms.
- To collect and analyze survey responses.
- To suggest practical methods for reducing energy wastage.

Secondary Objectives:

- To apply web development skills using HTML, CSS, and JavaScript.
- To improve teamwork and communication skills.
- To understand community energy consumption patterns.

- To analyze survey data using charts and graphs.
- To prepare a technical report based on the internship work.

1.3 Scope

This report describes all the activities carried out during the internship, including project planning, website development, survey preparation, data collection, survey analysis, findings, and recommendations related to energy conservation. The project mainly focuses on creating awareness and analyzing survey responses; it does not include hardware implementation or real-time monitoring of electricity consumption.

The survey was conducted for a limited period among a specific group of people; therefore, the results may not represent the opinions of the entire population. Even with this limitation, the study provides useful information about community energy usage and highlights simple methods that can help people conserve electricity in their daily lives.

1.4 Area (Location) and Demographics

The internship and community survey were carried out in and around Surat, Gujarat, covering respondents from the SSASIT college network, WhatsApp groups, and social media platforms. The survey reached a diverse demographic including students, working professionals, homemakers, and senior citizens, ensuring that the findings reflect a broad cross-section of the community rather than a single occupational or age group.

1.5 Infrastructure, Theme and Justification

The project theme selected was Energy Conservation, given its high relevance to Gujarat's rapidly growing electricity demand and the hot climate of Surat, which drives heavy cooling-related consumption. The infrastructure used for the project consisted of a self-developed website (hosted using standard web technologies), Google Forms for survey administration, and Google Sheets/Microsoft Excel for data storage and analysis.

The theme was justified by the observation that while government schemes such as UJALA and the Star Rating Programme have improved energy efficiency at a policy level, community-level awareness and behavioural change remain incomplete. This gap justified a student-led digital awareness initiative combining a web portal with an evidence-based community survey.

Chapter 2: Proposal

2.1 Problem Statement

Despite reasonable awareness of energy conservation concepts, households continue to waste significant electricity through excessive cooling appliance use, poor standby-power habits, incomplete transition to LED lighting, and low participation in awareness programmes. There is a need for an accessible digital platform that both educates the community and gathers first-hand data on local energy behaviour, so that targeted, practical recommendations can be made.

2.2 Stakeholders

- Community members / households (survey respondents and beneficiaries of awareness content).
- Gujarat Technological University (GTU) – programme owner of the Summer Internship.
- SSASIT, Surat, Computer Engineering Department – institutional guidance and mentorship.
- Faculty Guide, Prof. Pritesh Bhimani – academic supervision.
- Project team of seven Computer Engineering students – design, development, and analysis.
- Government bodies such as BEE and MNRE – indirect stakeholders through referenced schemes.

2.3 Methodology

The project followed a structured methodology consisting of problem identification, literature review, project planning, survey design, data collection, data analysis, website development, and preparation of recommendations.

- Project Planning: Selecting the topic, discussing ideas, and assigning individual responsibilities to team members for the fifteen-day internship period.
- Literature Review: Studying articles, government websites, and educational resources on energy conservation to inform the survey questionnaire and website content.

- Survey Design: Preparing an 11-question Google Form covering awareness, habits, and attitudes, using simple language accessible to varied age and educational groups.
- Data Collection: Sharing the survey link through online platforms and personal contacts, with responses recorded automatically in Google Sheets.
- Data Analysis: Analyzing responses using charts and graphs to identify usage patterns, awareness levels, and improvement areas.
- Website Development: Building the Smart Community Energy Portal using HTML, CSS, and JavaScript, with a responsive interface for mobile and desktop access.

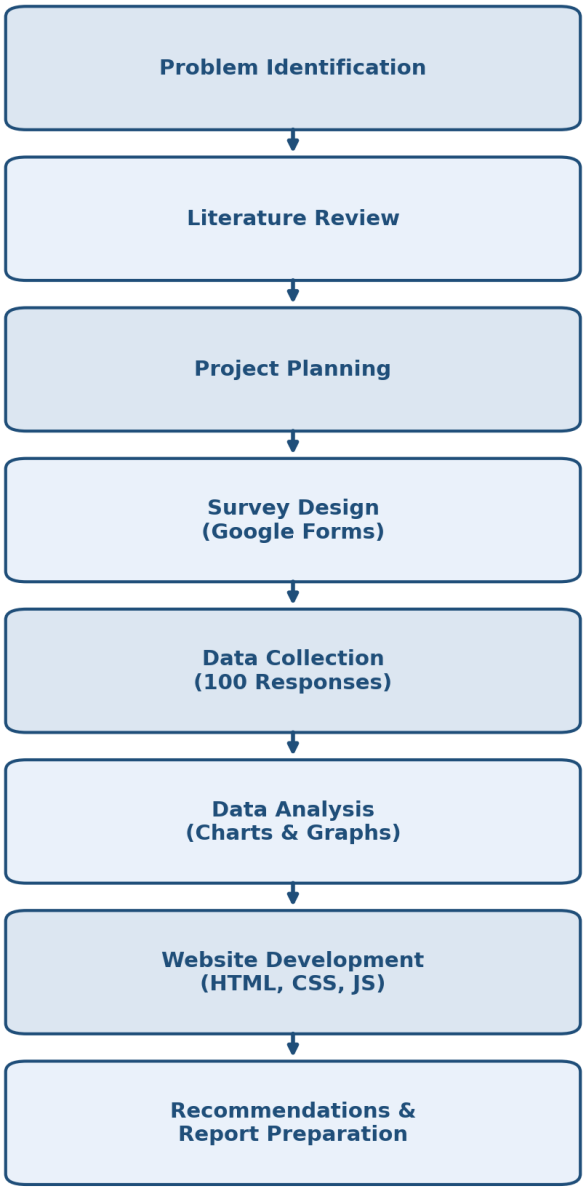


Figure 2.1: Project Methodology – Activity Flowchart

2.4 Timeline

Table 2.1: Project Implementation Timeline

Phase	Activity	Deliverable	Duration
Planning	Team formation, scope definition, tool setup	Project plan, role assignment	Day 1–2
Research	Literature review, content development, survey design	Research notes, website content, survey questionnaire	Day 2–4
Development	Website coding, testing, integration	Functional multi-page website	Day 3–9
Survey	Survey distribution, response monitoring, collection	100 survey responses	Day 5–12
Analysis & Reporting	Data analysis, visualization, report writing	Report, presentation, findings	Day 10–15

Chapter 3: Field Work

3.1 Survey Design and Distribution

The Community Energy Survey was designed as an 11-question Google Form covering key dimensions of household energy behaviour, awareness, and attitudes. It used a mix of multiple-choice questions, rating scales, and multi-select questions, and was distributed through WhatsApp groups, college networks, social media (Instagram and Facebook), and QR codes embedded in the Smart Community Energy Portal website.

The team actively encouraged participation and followed up to ensure diverse respondent demographics including students, working professionals, homemakers, and senior citizens. The survey successfully collected 100 responses over approximately 7 days of active distribution (approximately 19–27 June 2026).

Table 3.1: Survey Design Summary

Survey Parameter	Details
Survey Platform	Google Forms
Number of Questions	11
Total Responses Collected	100
Distribution Method	WhatsApp, Social Media, QR Code on Website
Survey Duration	Approximately 7 days
Question Types	Multiple Choice, Multi-select, Rating Scale (1–5)
Response Format	Anonymous
Analysis Tool	Google Forms built-in analytics + Manual analysis

3.2 Observations

Field observations during survey distribution indicated strong willingness among community members to participate once the purpose of the survey was explained. Respondents from the student community were quickest to respond via WhatsApp and social media, while homemakers and senior citizens were more often reached through personal contacts and the QR code placed on the website. This variation confirmed the value of using multiple distribution channels rather than a single platform.

3.3 Question-by-Question Survey Analysis (Q1–Q11)

The full survey questionnaire, response tables, and detailed analysis for all eleven questions are presented below, exactly as recorded from the 100 community responses.

Table 3.2: Survey Questionnaire Overview (Q1–Q11)

Q#	Question	Type
Q1	Do you think energy conservation can help reduce environmental pollution?	Single Choice
Q2	How often do you use natural light instead of electric lights during the daytime?	Single Choice
Q3	How many hours per day do you use electrical appliances?	Single Choice
Q4	Do you unplug chargers and devices when they are not in use?	Single Choice
Q5	Have you participated in any energy conservation awareness program?	Single Choice
Q6	Which type of lighting is mostly used in your home?	Single Choice
Q7	Have you noticed an increase in your electricity bill during the last year?	Single Choice
Q8	What are the main reasons for energy wastage in households?	Multi-select
Q9	Who should take the lead in spreading awareness about energy conservation?	Single Choice
Q10	What motivates people most to conserve energy?	Multi-select
Q11	Rate your awareness of energy conservation practices (1–5 scale)	Rating Scale

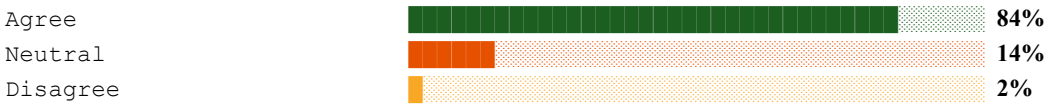
3.3.1 Question 1: Energy Conservation and Environmental Pollution

Survey Question: Do you think energy conservation can help reduce environmental pollution?

Table 3.3: Question 1 Response Distribution – Energy Conservation and Environmental Pollution

Response Option	Count	Percentage
Agree	84	84%
Neutral	14	14%
Disagree	2	2%
TOTAL	100	100%

Response Distribution:



This question measured the most fundamental awareness about the link between energy use and environmental pollution. An overwhelming 84% of respondents agree that energy conservation helps reduce environmental pollution, reflecting a high level of basic environmental consciousness in the community.

The 14% neutral responses suggest that while these individuals neither strongly disagree nor fully agree, they may lack the specific knowledge to form a firm position. Only 2% disagreed, which is reassuringly low but may require personalized engagement strategies.

Key Finding: Strong awareness of the energy-environment link provides an excellent foundation for community engagement. Campaigns can build on this awareness to motivate behavioural change, rather than spending resources on basic environmental education.

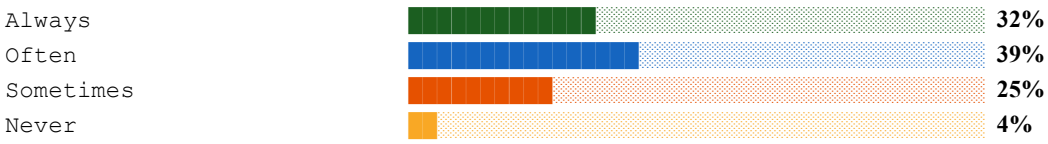
3.3.2 Question 2: Use of Natural Light

Survey Question: How often do you use natural light instead of electric lights during the daytime?

Table 3.4: Question 2 Response Distribution – Use of Natural Light

Response Option	Count	Percentage
Always	32	32%
Often	39	39%
Sometimes	25	25%
Never	4	4%
TOTAL	100	100%

Response Distribution:



A combined 71% of respondents (Always + Often) regularly use natural light during daytime, a positive indicator of energy-conscious behaviour. The 25% who only Sometimes use natural light may be influenced by room layout or working in interior spaces, while the 4% who Never use it may work in windowless spaces.

Key Finding: Natural light utilization is reasonably good, but the 29% who don't consistently use it represent an opportunity for awareness campaigns and building-design recommendations.

3.3.3 Question 3: Daily Electrical Appliance Usage

Survey Question: How many hours per day do you use electrical appliances?

Table 3.5: Question 3 Response Distribution – Daily Electrical Appliance Usage

Response Option	Count	Percentage
Less than 4 hours	12	12%
4-8 hours	33	33%
8-12 hours	37	37%
More than 12 hours	18	18%
TOTAL	100	100%

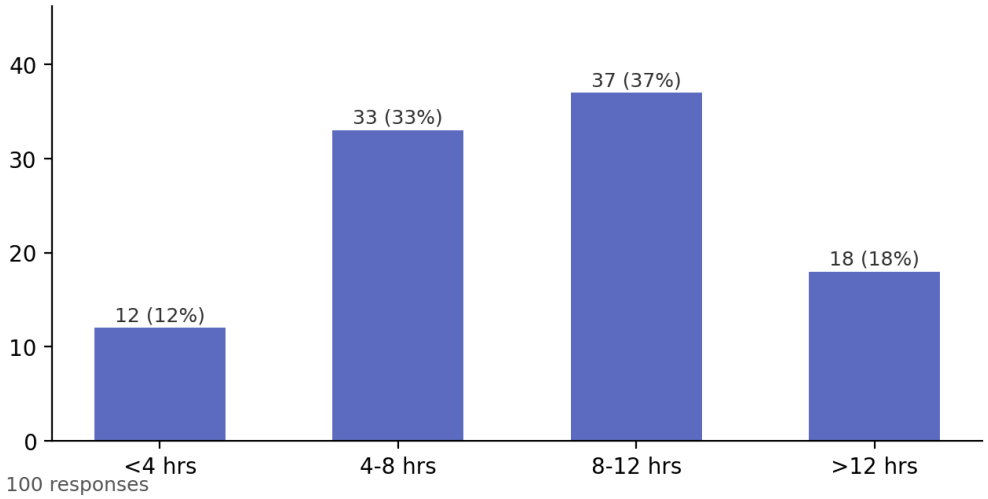


Figure 3.1: Question 3 – Daily Electrical Appliance Usage (Response Chart)

A significant 55% of respondents use electrical appliances for 8 or more hours per day, a high consumption level. The 18% who use appliances for more than 12 hours daily are at particularly high risk for elevated electricity bills and energy wastage. Only 12% use appliances for less than 4 hours daily.

Key Finding: High appliance usage hours are a significant contributor to household electricity bills. Recommendations around appliance efficiency, usage scheduling, and

standby-power reduction could meaningfully reduce consumption for the majority of respondents.

3.3.4 Question 4: Unplugging Devices When Not in Use

Survey Question: Do you unplug chargers and devices when they are not in use?

Table 3.6: Question 4 Response Distribution – Unplugging Devices When Not in Use

Response Option	Count	Percentage
Always	48	48%
Sometimes	24	24%
Rarely	20	20%
Never	8	8%
TOTAL	100	100%

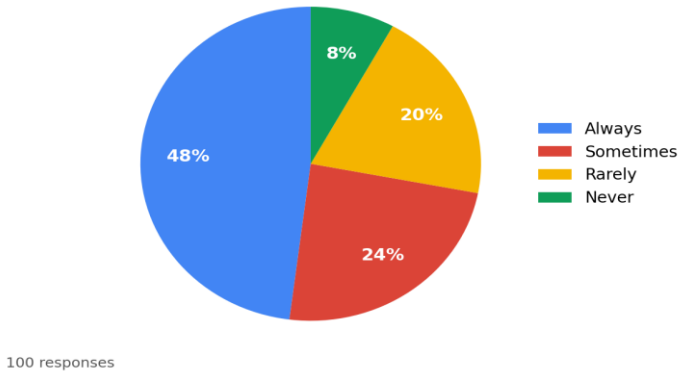


Figure 3.2: Question 4 – Unplugging Devices When Not in Use (Response Chart)

Unplugging devices avoids standby (phantom) power draw. While 48% always unplug devices, the remaining 52% leave devices plugged in at least sometimes, and this collective phantom load adds up to significant wasted electricity across households. The 28% who Rarely or Never unplug devices are the primary target for awareness campaigns on this issue.

Key Finding: Phantom power from always-plugged devices is an invisible but significant source of waste. Campaigns promoting the habit of unplugging, along with smart power strips, can address this issue.

3.3.5 Question 5: Participation in Awareness Programs

Survey Question: Have you participated in any energy conservation awareness program?

Table 3.7: Question 5 Response Distribution – Participation in Awareness Programs

Response Option	Count	Percentage
Yes	43	43%
No	57	57%
TOTAL	100	100%

Response Distribution:



More than half (57%) of respondents have never participated in an energy conservation awareness program, indicating a significant outreach gap. The 43% who have participated is higher than might be expected in the absence of a systematic community programme.

Key Finding: There is a significant awareness-program participation gap. Digital platforms, gamification, peer-to-peer sharing, and integration into school curricula can help increase participation rates.

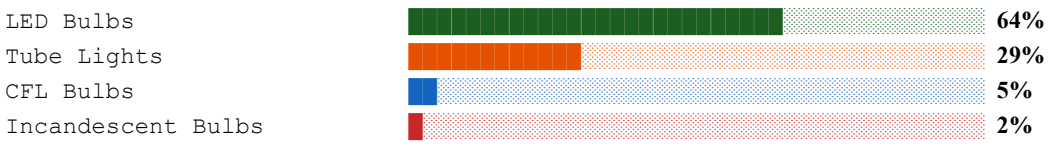
3.3.6 Question 6: Type of Lighting Used at Home

Survey Question: Which type of lighting is mostly used in your home?

Table 3.8: Question 6 Response Distribution – Type of Lighting Used at Home

Response Option	Count	Percentage
LED Bulbs	64	64%
CFL Bulbs	5	5%
Tube Lights	29	29%
Incandescent Bulbs	2	2%
TOTAL	100	100%

Response Distribution:



The dominance of LED bulbs at 64% reflects the success of India's UJALA scheme and general market transition to LED lighting. Tube lights at 29% remain a concern, since

conventional fluorescent tube lights are significantly less efficient than LED equivalents. CFL bulbs (5%) and incandescent bulbs (2%) represent smaller but still meaningful upgrade opportunities.

Key Finding: LED adoption is strong but not complete. A targeted campaign encouraging the remaining 36% of households to switch to LED alternatives could yield significant energy savings across the community.

3.3.7 Question 7: Increase in Electricity Bills

Survey Question: Have you noticed an increase in your electricity bill during the last year?

Table 3.9: Question 7 Response Distribution – Increase in Electricity Bills

Response Option	Count	Percentage
Yes	56	56%
No	23	23%
Maybe / Unsure	21	21%
TOTAL	100	100%

Response Distribution:



A majority of 56% of respondents have noticed an increase in their electricity bills over the past year, reflecting both tariff increases and potentially increased consumption. The 21% who are unsure may not be actively monitoring their bills, which itself indicates low energy awareness.

Key Finding: Rising electricity bills are a lived reality for the majority of the community and can serve as a powerful motivator for behavioural change through economically-framed messaging.

3.3.8 Question 8: Main Reasons for Energy Wastage

Survey Question: What are the main reasons for energy wastage in households? (Multiple answers permitted)

Table 3.10: Question 8 Response Distribution – Main Reasons for Energy Wastage

Response Option	Count	Percentage of Respondents
Excessive AC/Fan Usage	48	48%
Leaving Lights On	40	40%
Lack of Awareness	32	32%
Old Appliances	29	29%
Other	13	13%

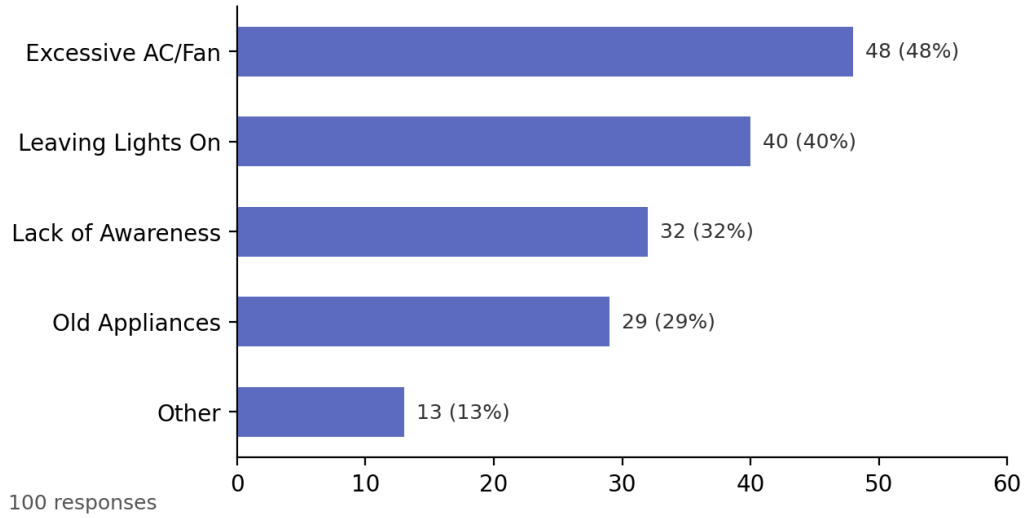


Figure 3.3: Question 8 – Main Reasons for Energy Wastage (Response Chart)

Excessive AC/Fan usage, cited by 48% of respondents, reflects the climate reality of Surat and Gujarat, where hot summers drive heavy cooling demand. Leaving lights on (40%) is a behavioural issue addressable through awareness and habit change. Lack of awareness (32%) validates the core premise of this project, while old appliances (29%) represent a technology-upgrade challenge.

Key Finding: A multi-pronged approach is needed: behavioural change campaigns targeting lights and cooling habits, appliance upgrade incentives, and expanded awareness programmes, with cooling behaviour as the top priority for the Surat climate context.

3.3.9 Question 9: Responsibility for Awareness

Survey Question: Who should take the lead in spreading awareness about energy conservation?

Table 3.11: Question 9 Response Distribution – Responsibility for Awareness

Response Option	Count	Percentage
Government	36	36%
Schools and Colleges	34	34%
Individuals	20	20%
Industries	10	10%
TOTAL	100	100%

Government (36%) and educational institutions (34%) are nearly equally favoured as the primary drivers, together accounting for 70% of responses. This aligns well with the approach of this project, conducted by a college to bridge the educational institutions' mandate. The 20% who cite individuals reflects a healthy sense of personal responsibility.

Key Finding: The community expects government and educational institutions to lead energy conservation awareness, validating the role of academic projects such as the Smart Community Energy Portal.

3.3.10 Question 10: Motivations for Energy Conservation

Survey Question: What motivates people most to conserve energy? (Multiple answers permitted)

Table 3.12: Question 10 Response Distribution – Motivations for Energy Conservation

Response Option	Count	Percentage
All factors combined	58	58%
Lower Electricity Bills	19	19%
Environmental Protection	17	17%
Government Rules	14	14%
Social Responsibility	13	13%

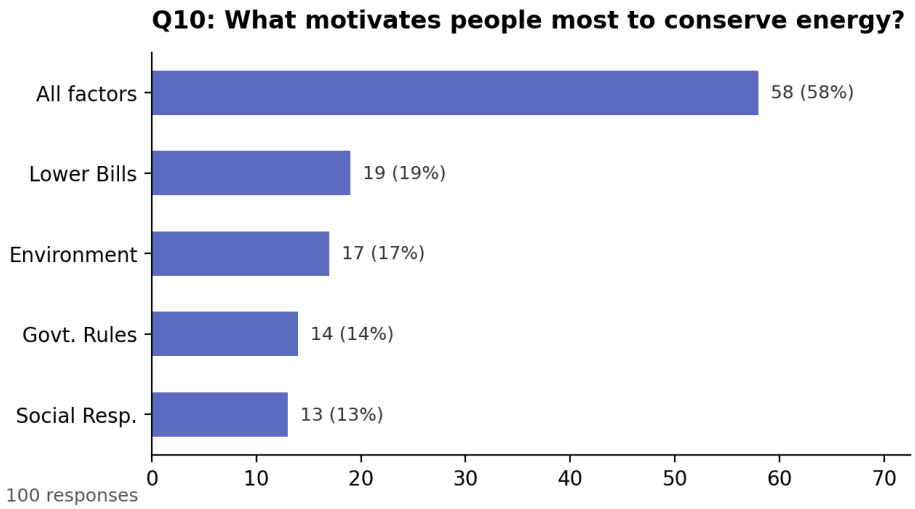


Figure 3.4: Question 10 – Motivations for Energy Conservation (Response Chart)

The dominant response of All factors combined (58%) indicates that people recognize energy conservation as a multi-faceted issue. Among single motivations, lower electricity bills (19%) leads, followed by environmental protection (17%), government rules (14%), and social responsibility (13%).

Key Finding: Since most people are motivated by the full spectrum of benefits, awareness campaigns should present energy conservation holistically rather than through single-issue messaging.

3.3.11 Question 11: Self-Rated Awareness Level

Survey Question: Rate your awareness of energy conservation practices on a scale of 1 (Very Low) to 5 (Very High).

Table 3.13: Question 11 Response Distribution – Self-Rated Awareness Level

Rating	Interpretation	Count	Percentage
1	Very Low Awareness	5	5%
2	Low Awareness	14	14%
3	Moderate Awareness	18	18%
4	Good Awareness	28	28%
5	Very High Awareness	35	35%

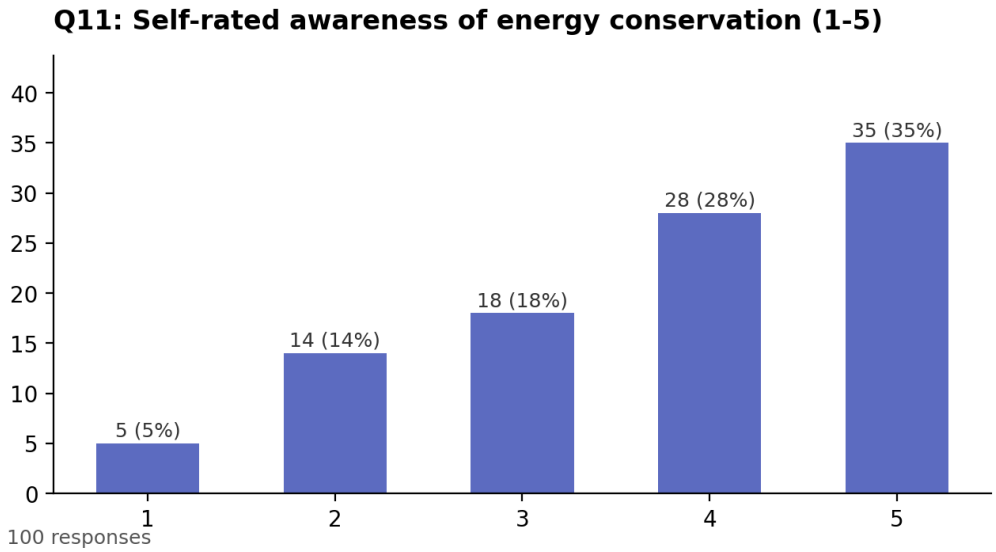


Figure 3.5: Question 11 – Self-Rated Awareness Level (Response Chart)

The awareness ratings show a positively skewed distribution with 63% of respondents rating themselves at Level 4 or 5. The mean awareness rating is approximately 3.74 out of 5. However, 19% rate themselves at Level 1 or 2, indicating genuine knowledge gaps. The disconnect between high self-rated awareness and suboptimal practices seen in earlier questions suggests awareness alone does not translate into behaviour change.

Key Finding: While community awareness is generally good (mean 3.74/5), the 19% with low self-rated awareness need targeted education, and the awareness-behaviour gap calls for practical, actionable guidance rather than information alone.

Chapter 4: Problem Analysis and Solution

4.1 Consolidated Findings

Based on the comprehensive analysis of all 11 survey questions, the following key findings were consolidated:

Table 4.1: Consolidated Findings

Finding Area	Key Data Point	Implication
Environmental Awareness	84% believe energy conservation reduces pollution	Strong foundation for engagement
Natural Light Usage	71% always/often use natural light	Target the remaining 29%
Appliance Usage Hours	55% use appliances 8+ hours/day	Appliance efficiency is critical
Device Unplugging	48% always unplug; 28% rarely/never do	Phantom power waste is significant
Program Participation	57% never participated in awareness programs	Large untapped audience
LED Adoption	64% use LED bulbs	Target the remaining 36%
Rising Bills	56% noticed bill increases	Economic motivation is strong
Main Waste Cause	48% cite excessive AC/Fan usage	Cooling behaviour is top priority
Responsibility	70% say Government/Schools should lead	Institutions have legitimacy
Motivation	58% motivated by all factors	Holistic messaging most effective
Self-Awareness	63% rate themselves Level 4–5; 19% at 1–2	Gap between self-rating and behaviour

4.2 Root Cause Analysis

Root Cause 1 – Excessive Cooling Energy Use: In the Surat climate, cooling loads are extremely high, particularly in summer. Inefficient AC/fan use (very low temperature settings, running units in unoccupied rooms, old inefficient equipment) is the single largest identified cause of household energy wastage (48%).

Root Cause 2 – Behavioural Energy Waste: Leaving lights on (40%) and not unplugging devices (52% sometimes or never do) reflect habitual, often unconscious behaviour patterns rather than intentional wastage, making them addressable through habit-formation campaigns.

Root Cause 3 – Incomplete LED Transition: 36% of households still use less efficient tube lights, CFLs, or incandescent bulbs, representing an unrealized efficiency opportunity.

Root Cause 4 – Awareness Program Gap: 57% of respondents have never participated in an awareness programme, showing that existing outreach mechanisms are insufficient.

Root Cause 5 – Awareness-Behaviour Gap: Despite high self-rated awareness (63% at Level 4–5), many respondents still show suboptimal behaviour, indicating that knowing is not the same as doing.

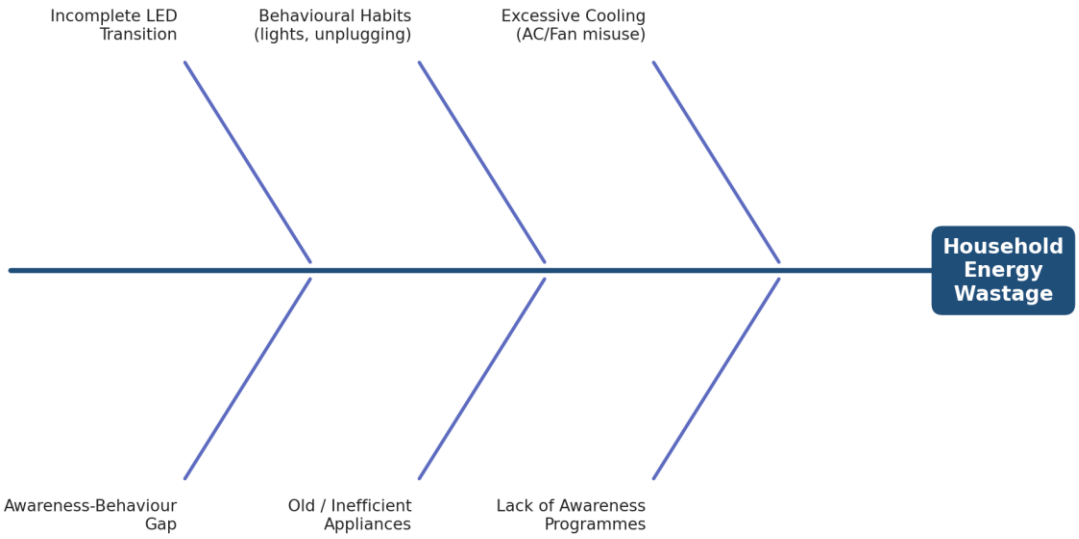


Figure 4.1: Fishbone (Ishikawa) Diagram – Root Causes of Household Energy Wastage

4.3 SWOT Analysis

Table 4.2: SWOT Analysis

Category	Details
Strengths	High baseline environmental awareness (84%); strong LED adoption (64%); good natural-light usage (71%); college-led credibility for awareness initiatives.
Weaknesses	High appliance usage hours (55% at 8+ hrs/day); low unplugging discipline in 52% of households; low awareness-programme participation (57% never participated).
Opportunities	70% of respondents want government/schools to lead — institutions can scale awareness; smart plugs and LED-tube retrofits offer quick wins; digital platforms can reach the 57% untapped audience.
Threats	Rising electricity bills without matching behaviour change could increase household financial stress; continued use of old, inefficient appliances; climate-driven cooling demand likely to keep growing.

4.4 Empathy Mapping

A brief empathy exercise, informed by the survey responses, was used to understand the community's perspective:

- Says: “I know energy conservation is important, but I don't always have time to think about it.”
- Thinks: Believes conservation matters environmentally, but is unsure which specific actions make the biggest difference.
- Feels: Slightly guilty about rising bills, but motivated primarily by cost savings rather than compliance.
- Does: Uses LED bulbs and natural light reasonably well, but leaves devices plugged in and runs AC/fans for long hours out of comfort or habit.

4.5 Prototype / Solution

The Smart Community Energy Portal itself is the project's prototype solution — a website built using HTML, CSS, and JavaScript that provides energy-conservation information, practical energy-saving tips, renewable versus non-renewable energy education, and an embedded community survey (with QR code access). It was designed with a simple, responsive interface accessible on both mobile phones and computers, directly targeting the awareness-programme gap identified in the survey.

Building on this prototype, the following targeted solutions are proposed for each root cause identified above:

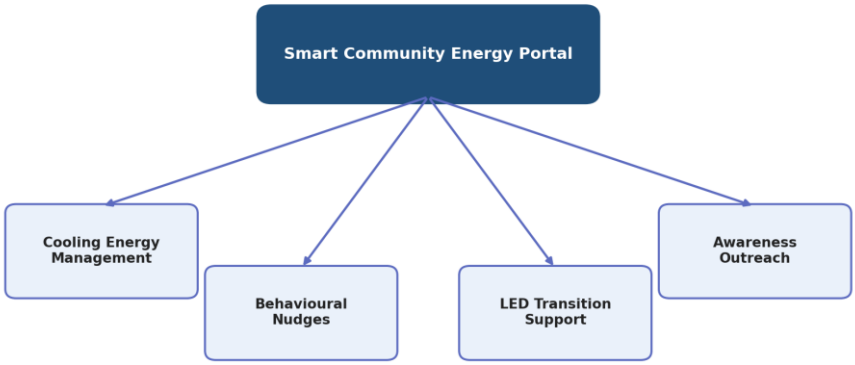


Figure 4.2: Solution Framework Derived from the Smart Community Energy Portal

Solution for Excessive Cooling Energy Use:

- Set air conditioners to 24–26°C, which is comfortable and significantly more efficient than typical 18–20°C settings.
- Use ceiling fans alongside AC to allow the thermostat to be set 2–3 degrees higher.
- Turn off AC and fans when leaving a room, even briefly.
- Use curtains, blinds, or reflective window films to reduce solar heat gain.
- Replace ACs older than 5–7 years with 5-star inverter units (30–40% less electricity).
- Community-level awareness campaigns on the 24–26°C setting, plus subsidy programmes for inverter AC upgrades.

Solution for Behavioural Energy Waste:

- "Turn Off All" visual reminders near switches and power strips; a "last one out" household rule.
- Smart power strips and motion-sensor lights to eliminate phantom loads without requiring behaviour change.
- Monthly/weekly energy-tracking challenges within communities or apartment complexes.
- School programmes teaching children energy-saving habits to bring home.

Solution for Incomplete LED Transition:

- Awareness campaigns targeted at tube-light users on comparably priced LED tube replacements (30–50% savings).
- Community bulk-procurement of LED products at discounted prices.
- Promotion of UJALA and related schemes for subsidized LED distribution.
- Proper disposal programmes for old tube lights and CFLs containing mercury.

Solution for Awareness Program Gap:

- Providing reliable, centralized information about energy conservation through the portal.
- Encouraging participation in community surveys.
- Sharing practical, actionable energy-saving tips.
- Increasing awareness through educational content and interactive webpages.
- Helping users understand sustainable energy through visual learning.

4.6 Feasibility, Advantages and Limitations

Feasibility: All proposed solutions use existing, low-cost technologies (LED bulbs, smart plugs, awareness content) and require no specialized hardware development, making them feasible to implement at the household and community level without significant investment.

Advantages: The recommendations directly target the specific causes of wastage identified in the survey data (cooling behaviour, standby power, lighting choice, and awareness gaps),

so they are evidence-based rather than generic. They also leverage the community's stated preference for government/school-led initiatives, improving the likelihood of adoption.

Limitations: The survey sample of 100 respondents, while diverse, was collected over a short period and may not represent the entire population of Surat or Gujarat. The project addresses awareness and behaviour but does not include hardware-based real-time energy monitoring, and its impact on actual electricity consumption has not been independently measured.

Chapter 5: Implementation Plan

5.1 Resources

The following tools and technologies were used to implement the project:

Table 5.1: Tools and Technologies Used

Tool/Technology	Purpose
HTML	Structure of the website
CSS	Website design and styling
JavaScript	Interactive features
Google Forms	Community survey
Google Sheets	Data collection and analysis
Microsoft Word	Report preparation
Microsoft Excel	Charts and graphical analysis

Human resources consisted of a team of seven Computer Engineering students with individually assigned responsibilities (website development, survey design, distribution, data analysis, and documentation), guided by Prof. Pritesh Bhimani.

5.2 Budget

As the project relied entirely on free-tier digital tools (Google Forms, Google Sheets) and open web technologies (HTML, CSS, JavaScript) hosted without paid infrastructure, no significant monetary budget was required. The primary investment was the team's time over the fifteen-day internship period. For future scaling (e.g., a mobile application or paid hosting/domain), a modest budget would be required, as noted under Future Scope.

5.3 Schedule

Table 5.2: Project Implementation Schedule

Phase	Activity	Deliverable	Duration
Planning	Team formation, scope definition, tool setup	Project plan, role assignment	Day 1–2
Research	Literature review, content development, survey design	Research notes, website content, survey questionnaire	Day 2–4
Development	Website coding, testing, integration	Functional multi-page website	Day 3–9
Survey	Survey distribution, response monitoring, collection	100 survey responses	Day 5–12
Analysis & Reporting	Data analysis, visualization, report writing	Report, presentation, findings	Day 10–15

Chapter 6: Impact Assessment

6.1 Social Impact

The project increased community engagement around energy conservation, with 100 households or individuals directly participating in the survey and being exposed to awareness content through the portal. By involving students, working professionals, homemakers, and senior citizens, the initiative encouraged cross-generational conversation about responsible electricity usage and demonstrated that educational institutions can meaningfully lead community awareness efforts, a role that 70% of surveyed respondents said institutions and government should take on.

6.2 Economic Impact

With 56% of respondents already reporting increased electricity bills, the recommendations on AC temperature settings, appliance upgrades, and phantom-load reduction have direct potential to lower household electricity expenditure if adopted. Since 19% of respondents cited lower electricity bills as their key personal motivator, the economically-framed recommendations in Chapter 4 are well aligned with community priorities and can produce measurable savings without additional cost to households.

6.3 Environmental Impact

Reduced electricity consumption directly lowers demand on fossil-fuel-based power generation, thereby reducing greenhouse gas emissions and air pollution. Given that 84% of respondents already recognize this environmental link, the awareness content reinforces and channels existing environmental consciousness into concrete action. Wider LED adoption, reduced cooling wastage, and lower standby power draw would collectively contribute, even at a small community scale, to reduced carbon footprint and more sustainable use of natural resources.

Conclusion

The Smart Community Energy Portal internship project was successfully completed by seven Computer Engineering students of SSASIT, Surat, during the 15-day internship (15 June to 30 June 2026). The project achieved its objective of promoting energy conservation awareness through an interactive digital platform.

The community survey, consisting of 100 responses, provided valuable insights into people's energy consumption habits, awareness levels, and conservation practices. The results showed that while many participants understand the importance of energy conservation and use energy-efficient methods such as LED lighting and natural daylight, challenges remain, including excessive use of air conditioners and fans, behavioural energy wastage, and limited participation in awareness programmes.

The developed portal addresses these challenges by providing information about renewable and non-renewable energy sources, community survey analysis, data visualizations, energy-saving tips, and educational resources. It encourages community participation and promotes sustainable energy practices through digital awareness. The project also helped the team gain practical experience in web development, survey design, data analysis, teamwork, and project management.

In conclusion, the Smart Community Energy Portal demonstrates how digital technology can support public awareness and encourage responsible energy conservation. With future enhancements and regular updates, the portal can continue to serve as a useful educational resource for promoting sustainable energy practices in the community.

Future Scope

The Smart Community Energy Portal has successfully achieved its objective of promoting energy conservation awareness through digital technology. In the future, the portal can be enhanced by introducing additional interactive features and expanding its reach to a larger community:

- Develop a mobile application to provide easy access to the portal anytime and anywhere.

- Add an energy consumption calculator to help users estimate household electricity usage.
- Introduce AI-based personalized energy-saving recommendations based on user inputs.
- Provide multilingual support (English, Hindi, and Gujarati) to reach a wider audience.
- Expand the community survey and awareness resources with more educational content, interactive quizzes, and updated energy conservation information.

References

Government and Regulatory Sources

- Bureau of Energy Efficiency (BEE), Ministry of Power, Government of India. Energy Conservation and Energy Efficiency Resources.

  Available at: https://www.beeindia.gov.in

- Ministry of New and Renewable Energy (MNRE), Government of India. Renewable Energy Information and Policies.

  Available at: https://mnre.gov.in

Project-Specific References

- Smart Community Energy Portal – Internship project website developed by the Computer Engineering students of SSASIT, Surat.
- Community Energy Conservation Awareness Survey – Online survey conducted using Google Forms with 100 community responses during the internship.
- Survey Response Dashboard – Data visualization generated from Google Forms survey responses.
- GTU Summer Internship Programme Guidelines (2025–26) – Gujarat Technological University, Ahmedabad.
- Department of Computer Engineering, SSASIT, Surat – Project guidance, internship activities, and academic support.

--- End of Report ---

Smart Community Energy Portal

GTU Internship Project | SSASIT, Surat | June 2026

"Together, we can build a smarter and greener community."